11

Why Is There Space?

Emergence and Locality

Okay, at long last we're ready to think about the actual world.

Wait a minute, I hear you thinking. *I thought we were talking about the actual world already. Isn't quantum mechanics supposed to describe the actual world?*

Well, sure. But quantum mechanics can also describe plenty of worlds other than our actual one. Quantum mechanics itself isn't a single theory, in the sense of being a model of one specific physical system. It's a framework, just like classical mechanics is, in which we can talk about many different physical systems. We can talk about the quantum theory of a single particle, or of the electromagnetic field, or of a set of spins, or of the entire universe. Now it's time to focus in on what the quantum theory of our actual world might look like.

This goal—finding the right quantum theory of the actual world—has been pursued by generations of physicists since the early twentieth century. By any possible measure, they have been extraordinarily successful. One important insight was to think of the basic building blocks of nature not as particles but as fields pervading space, thus leading to *quantum field theory*.

Back in the nineteenth century, physicists seemed to be homing in on a view of the world in which both particles and fields played a role: matter was made of particles, and the forces by which they interacted were described by fields. These days we know better; even the particles that we know and love are actually vibrations in fields that suffuse the space around us. When we see particle-like tracks in a physics experiment, that's a